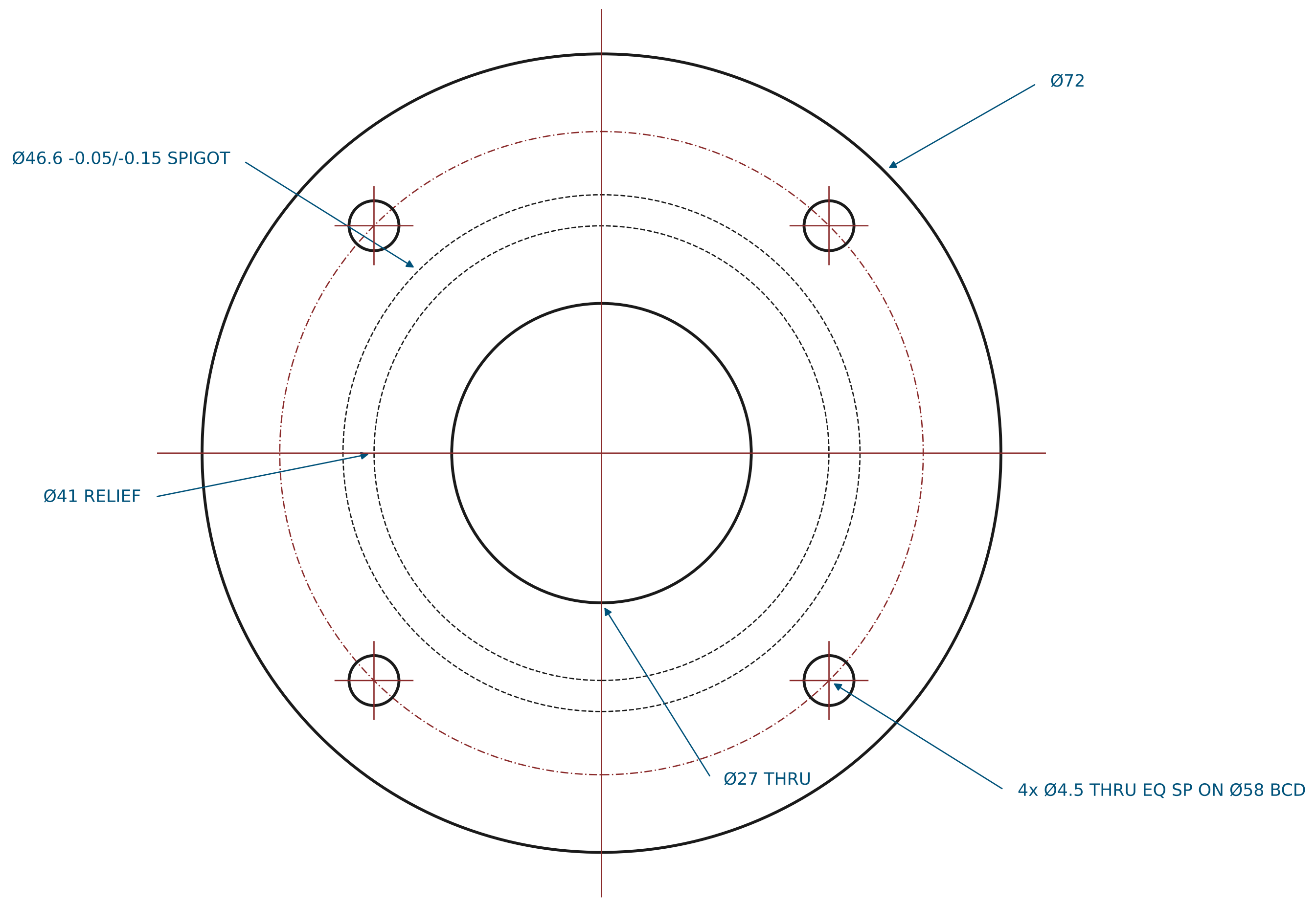
SHEET 1: BEARING RETAINER CAP   (BRG-CAP)

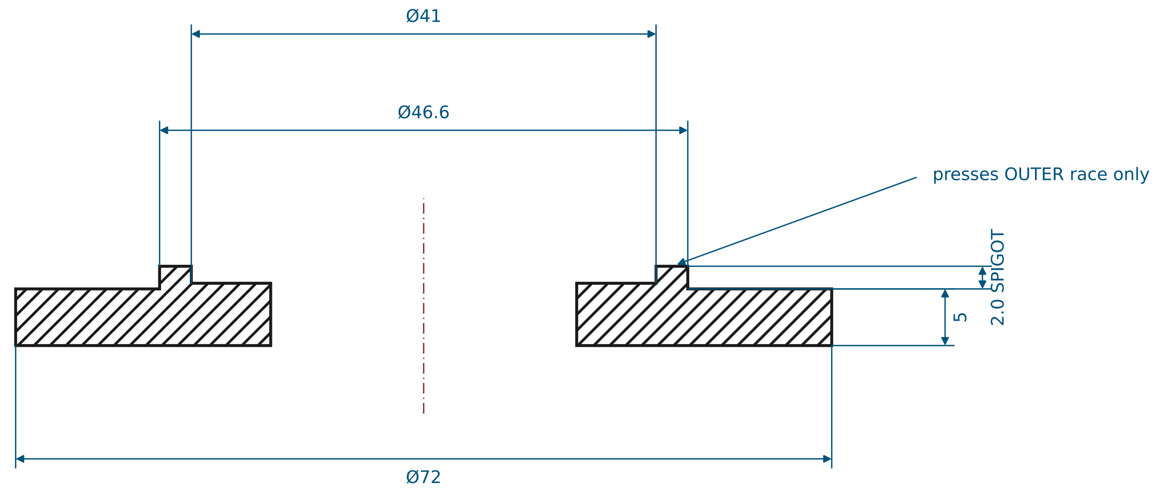
GENERAL TOLERANCES UNLESS NOTED: X ±0.5   X.X ±0.2   HOLES +0.2/-0   ANGLES ±0.5°   BREAK ALL EDGES 0.3

DUAL MATERIAL CALLOUTS: select ONE per order — printed parts take brass heat-set inserts at every thread callout; 6061 parts are tapped.

1. One cap above AND one below each 7005 bearing bore; the pair bolts together THROUGH the plate with 4x M4x35 SHCS + nyloc (plate holes 4.5 on the same BCD; grip 26 at the deck, 20 at the plate).
2. The 46.6 spigot registers in the 47.0 plate bore. Its 41.0-46.6 face ring presses the bearing OUTER race only; the bore is relieved inboard of 41.0 to clear the rotating inner race.
3. Build one cap spigot-UP and one spigot-DOWN per bore (same part, flipped).
4. OD is 72 so the bolt holes keep a full 1.5d rim (edge distance 4.75).



PLAN (spigot side away)



SECTION A-A

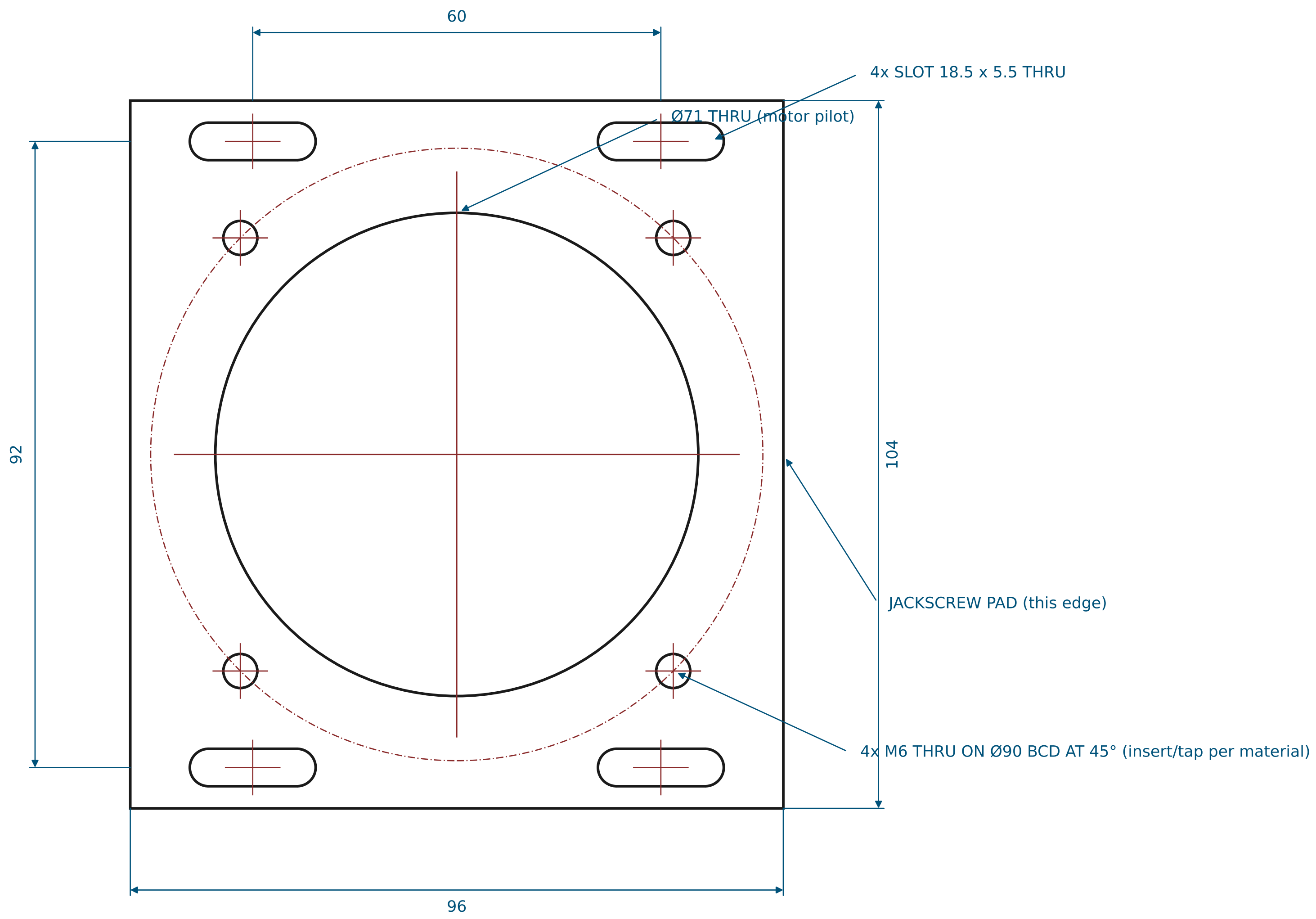
5-BAR BUNG-COVER ROBOT — BASE HARDWARE	
TITLE: BEARING RETAINER CAP	DWG NO: BCR-01 REV A
MATERIAL: PA12 MJF / 6061-T6	QTY: 8
SCALE 1.5:1 UNITS mm THIRD ANGLE	SHEET 1 OF 6 2026-07-19

SHEET 2: SLIDING MOTOR CARRIAGE (CARRIAGE)

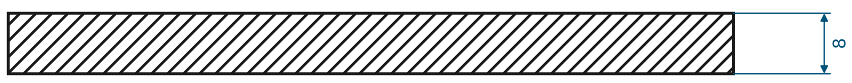
GENERAL TOLERANCES UNLESS NOTED: X ±0.5 X.X ±0.2 HOLES +0.2/-0 ANGLES ±0.5° BREAK ALL EDGES 0.3

DUAL MATERIAL CALLOUTS: select ONE per order — printed parts take brass heat-set inserts at every thread callout; 6061 parts are tapped.

1. Motor (A6M80, 80 sq flange) mounts from BELOW: 4x M6x18 SHCS up through the motor flange into the four M6 positions. PRINTED part: M6 brass heat-set inserts. 6061: tap M6.
2. The four slots take M5x16 SHCS + washer down into the cradle frame bosses directly beneath; slot length gives ±6 mm of tension travel.
3. The +X edge (toward the shoulder) is the jackscrew pad — the M6 jackscrew tip bears mid-edge.
4. Same part both sides (symmetric).



PLAN (top face)



EDGE VIEW

5-BAR BUNG-COVER ROBOT — BASE HARDWARE	
TITLE: SLIDING MOTOR CARRIAGE	DWG NO: BCR-02 REV A
MATERIAL: PA12 MJF / 6061-T6	QTY: 2
SCALE 1:1 UNITS mm THIRD ANGLE	SHEET 2 OF 6 2026-07-19

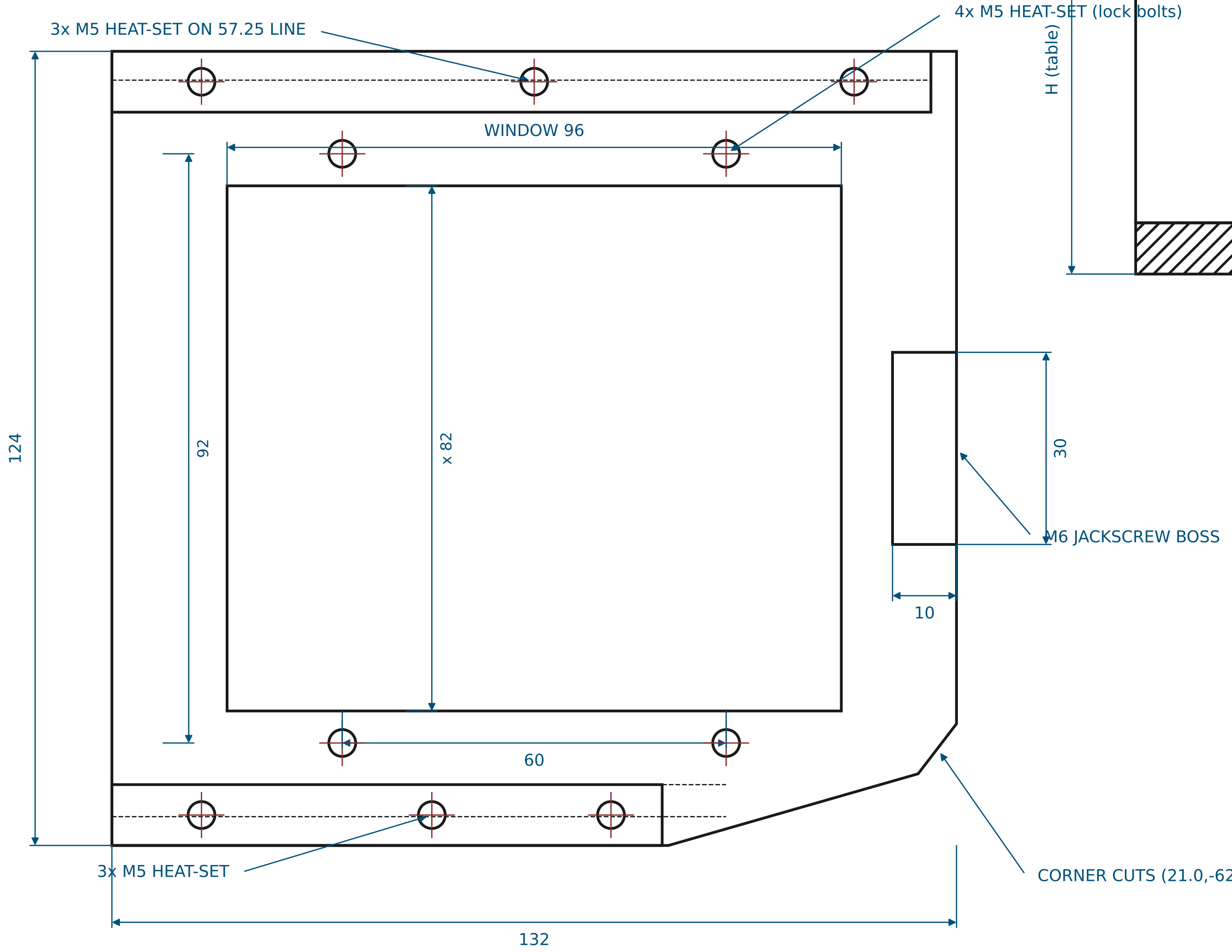
SHEET 3: SLIDER CRADLE (LH SHOWN, RH = MIRROR)   (CRADLE-L / CRADLE-R)

GENERAL TOLERANCES UNLESS NOTED: X ±0.5   X.X ±0.2   HOLES +0.2/-0   ANGLES ±0.5°   BREAK ALL EDGES 0.3

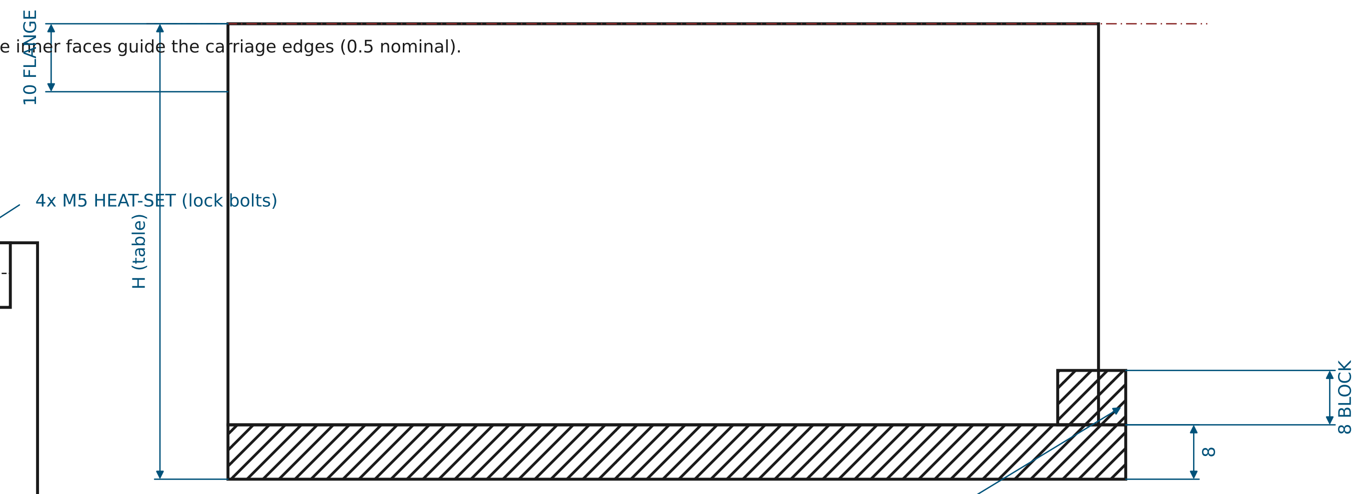
DUAL MATERIAL CALLOUTS: select ONE per order — printed parts take brass heat-set inserts at every thread callout; 6061 parts are tapped.

1. ONE printed part per motor = window frame + two shear fins + jack block. The RH part is the exact MIRROR of the LH drawn here; fin/wall heights differ per the table (belt planes stagger).
2. Fin top flanges (10 thick - full-depth M5 heat-set) bolt UP into the deck underside: 3x M5 per fin, M5x20 SHCS from the deck top (deck holes 5.2 THRU; ~8 mm engagement).
3. The four M5 positions in the frame border take the carriage lock bolts — fit heat-set inserts.
4. M6 heat-set (or tapped) jackscrew boss in the block; jam nut on the screw against the block face.
5. The chamfered corner keeps the two cradles clear of the machine centreline — do not 'square it up'.
6. FIN WALLS: inner faces at b = ±52.5, 5 thick, rising to the deck underside. Outboard wall spans a = -66..+62, inboard a = -66..+30. The inner faces guide the carriage edges (0.5 nominal).
7. Install the belt around both pulleys BEFORE bolting the deck down (the fins close the corridor).

PLAN — LH PART (frame top; fins rise toward viewer)



DATUM: origin at frame centre. +a = toward the shoulder (jack-block end), +b = toward the OUTBOARD fin. All holes in the HOLE TABLE.



ELEVATION (outboard fin)

PART TABLE

	FRAME TOP z	WALL H (frame btm→deck)	MIRROR
CRADLE-L	-7	67	as drawn
CRADLE-R	13	47	mirrored

HOLE TABLE (a, b) — LH part; RH = same with b negated

ID	a	b	SPEC
H1	-52.00	57.25	M5 heat-set, fin flange
H2	0.00	57.25	M5 heat-set, fin flange
H3	50.00	57.25	M5 heat-set, fin flange
H4	-52.00	-57.25	M5 heat-set, fin flange
H5	-16.00	-57.25	M5 heat-set, fin flange
H6	12.00	-57.25	M5 heat-set, fin flange
L1	30.00	46.00	M5 heat-set, carriage lock
L2	30.00	-46.00	M5 heat-set, carriage lock
L3	-30.00	46.00	M5 heat-set, carriage lock
L4	-30.00	-46.00	M5 heat-set, carriage lock
J1	61.00	0.00	M6 heat-set, jackscrew

5-BAR BUNG-COVER ROBOT — BASE HARDWARE	
TITLE: SLIDER CRADLE (LH SHOWN, RH = MIRROR)	DWG NO: BCR-03 REV A
MATERIAL: PA12 MJF / PETG-CF (print flat, frame down)	QTY: 1 + 1 (mirror)
SCALE 1:2 / 1:1 UNITS mm THIRD ANGLE	SHEET 3 OF 6 2026-07-19

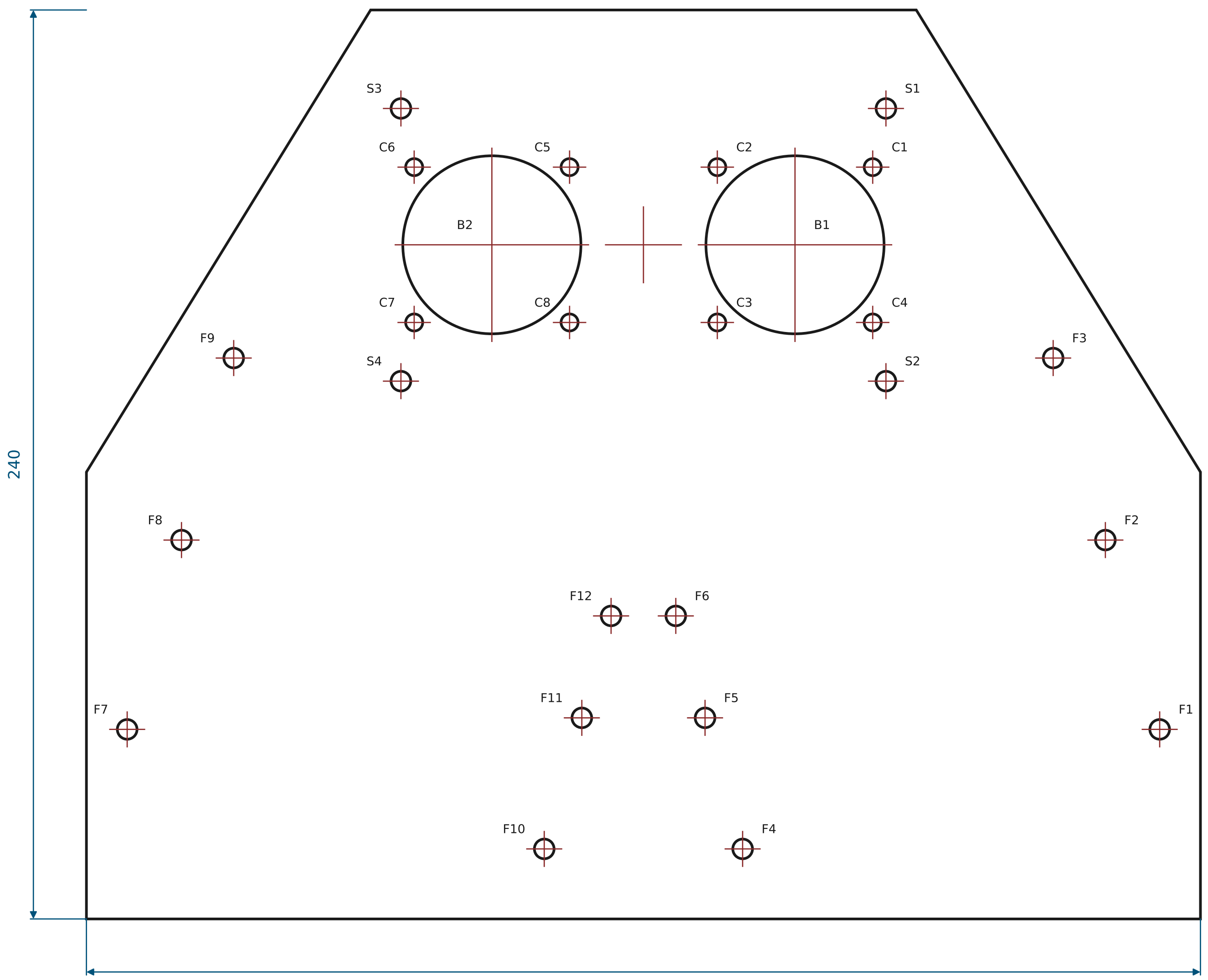
SHEET 4: BOTTOM DECK PLATE (DECK)

GENERAL TOLERANCES UNLESS NOTED: X ±0.5 X.X ±0.2 HOLES +0.2/-0 ANGLES ±0.5° BREAK ALL EDGES 0.3

DUAL MATERIAL CALLOUTS: select ONE per order — printed parts take brass heat-set inserts at every thread callout; 6061 parts are tapped.

1. Machine per the HOLE TABLE (coordinates from the plate origin = mid-span between the two shoulder bores; +Y toward the arms). All holes THRU.
2. The two 47.0 bores carry the 7005 bearings (transition fit: bore 47.0 +0.025/0).
3. Corner radii R8. Outline vertices in the VERTEX TABLE.
4. Plate must stay within a 300 x 300 print bed (current 294 x 240).

PLAN — deck top (origin at mid-span of shoulder bores)



HOLE TABLE

ID	X	Y	Ø	SPEC
B1	40.00	0.00	47.0	7005 bearing bore, thru
B2	-40.00	0.00	47.0	7005 bearing bore, thru
C1	60.51	20.51	4.5	M4 cap bolt, thru
C2	19.49	20.51	4.5	M4 cap bolt, thru
C3	19.49	-20.51	4.5	M4 cap bolt, thru
C4	60.51	-20.51	4.5	M4 cap bolt, thru
C5	-19.49	20.51	4.5	M4 cap bolt, thru
C6	-60.51	20.51	4.5	M4 cap bolt, thru
C7	-60.51	-20.51	4.5	M4 cap bolt, thru
C8	-19.49	-20.51	4.5	M4 cap bolt, thru
S1	64.00	36.00	5.2	M5 standoff bolt, thru
S2	64.00	-36.00	5.2	M5 standoff bolt, thru
S3	-64.00	36.00	5.2	M5 standoff bolt, thru
S4	-64.00	-36.00	5.2	M5 standoff bolt, thru
F1	136.25	-127.95	5.2	M5 fin bolt, thru
F2	121.91	-77.96	5.2	M5 fin bolt, thru
F3	108.13	-29.90	5.2	M5 fin bolt, thru
F4	26.18	-159.51	5.2	M5 fin bolt, thru
F5	16.26	-124.90	5.2	M5 fin bolt, thru
F6	8.54	-97.99	5.2	M5 fin bolt, thru
F7	-136.25	-127.95	5.2	M5 fin bolt, thru
F8	-121.91	-77.96	5.2	M5 fin bolt, thru
F9	-108.13	-29.90	5.2	M5 fin bolt, thru
F10	-26.18	-159.51	5.2	M5 fin bolt, thru
F11	-16.26	-124.90	5.2	M5 fin bolt, thru
F12	-8.54	-97.99	5.2	M5 fin bolt, thru

VERTEX TABLE (outline, R8 corners)

V1 (-72,62)	V2 (72,62)	V3 (147,-60)
V4 (147,-178)	V5 (-147,-178)	V6 (-147,-60)

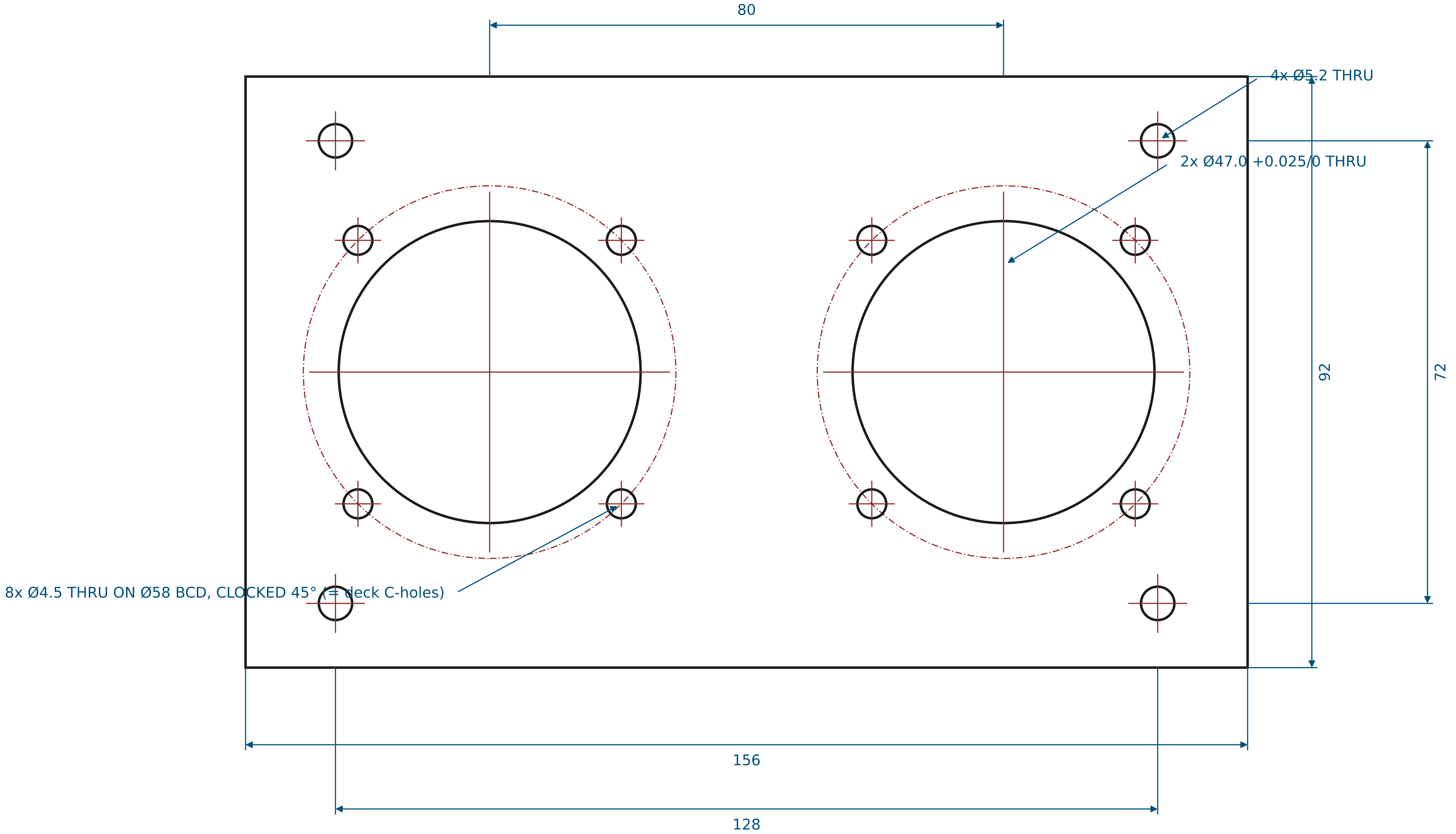
5-BAR BUNG-COVER ROBOT — BASE HARDWARE	
TITLE: BOTTOM DECK PLATE	DWG NO: BCR-04 REV A
MATERIAL: PA12 MJF / 6061-T6 (12 THK)	QTY: 1
SCALE 1:2 UNITS mm THIRD ANGLE	SHEET 4 OF 6 2026-07-19

SHEET 5: TOP PLATE + STANDOFFS   (TOP-PLATE)

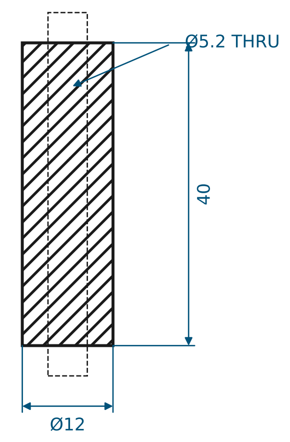
GENERAL TOLERANCES UNLESS NOTED: X ±0.5   X.X ±0.2   HOLES +0.2/-0   ANGLES ±0.5°   BREAK ALL EDGES 0.3

DUAL MATERIAL CALLOUTS: select ONE per order — printed parts take brass heat-set inserts at every thread callout; 6061 parts are tapped.

1. Same bearing bore + cap-bolt pattern as the deck (47.0 bores, 4x 4.5 on Ø58 BCD per bore).
2. Standoff bolt holes moved OUT to (±64, ±36) so the Ø72 caps clear the standoff bosses (they collided with the old Ø62 caps at ±60,±32).
3. 4x standoffs Ø12 x 40, Ø5.2 clearance bore THRU. Each stack is THROUGH-BOLTED: one M5 x 70 SHCS from the plate top, nyloc + washer under the deck (no threads in the standoff).



PLAN



STANDOFF (x4)

5-BAR BUNG-COVER ROBOT — BASE HARDWARE	
TITLE: TOP PLATE + STANDOFFS	DWG NO: BCR-05 REV A
MATERIAL: PA12 MJF / 6061-T6 (10 THK)	QTY: 1 (+4 standoffs)
SCALE 1:1 UNITS mm THIRD ANGLE	SHEET 5 OF 6 2026-07-19

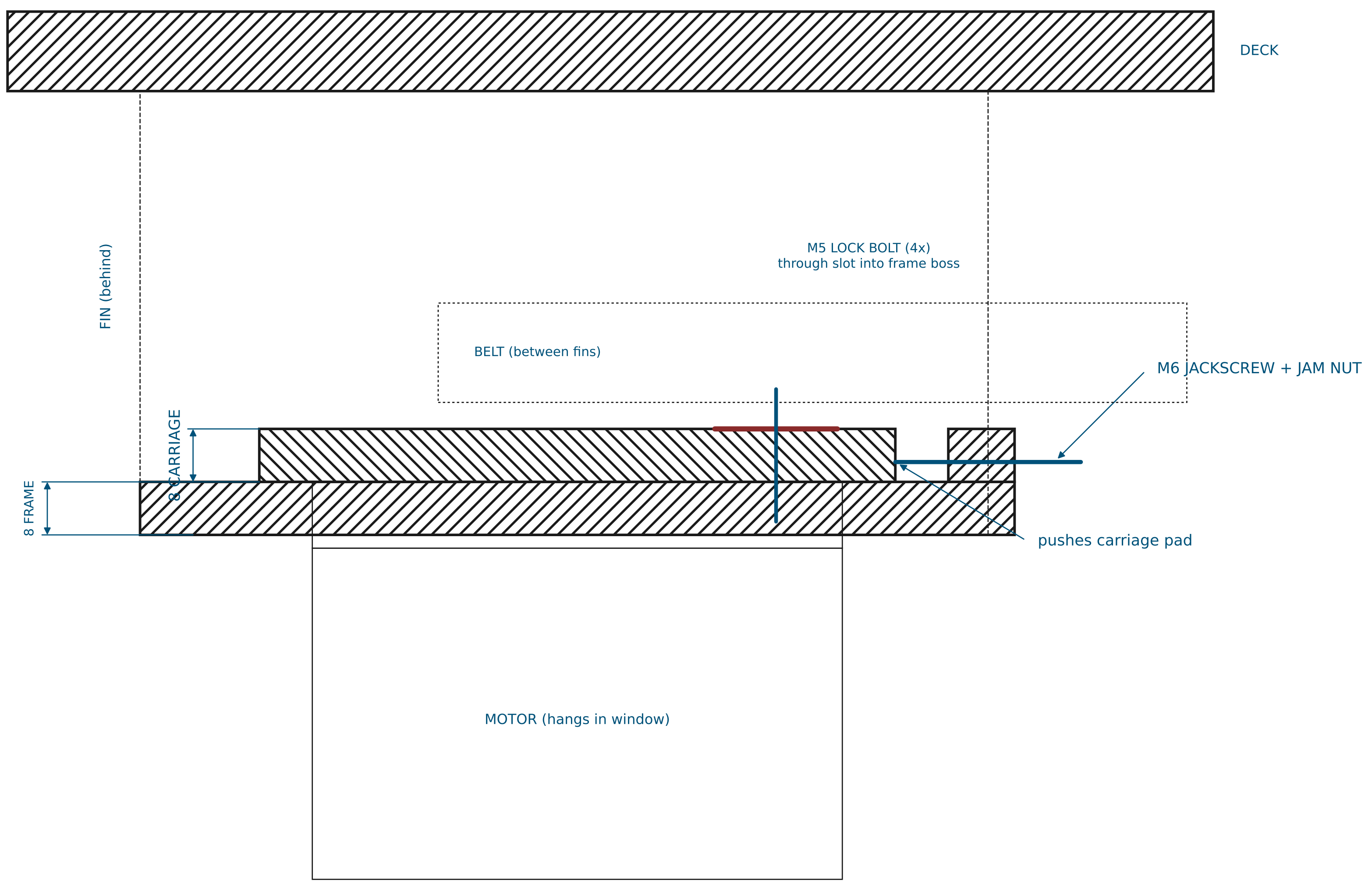
SHEET 6: TENSIONER ASSEMBLY + HARDWARE (ASSY-TENSIONER)

GENERAL TOLERANCES UNLESS NOTED: X ±0.5 X.X ±0.2 HOLES +0.2/-0 ANGLES ±0.5° BREAK ALL EDGES 0.3

DUAL MATERIAL CALLOUTS: select ONE per order — printed parts take brass heat-set inserts at every thread callout; 6061 parts are tapped.

1. TENSION PROCEDURE: loosen the 4x M5 lock bolts 1/2 turn -> turn the M6 jackscrew CW to push the carriage away from the shoulder (belt tightens) -> torque the lock bolts (M5: 4 N·m in inserts) -> set the M6 jam nut against the block.
2. Travel available ±6 mm. Belt: 450-5M-15 (24T -> 72T, C = 97.5 nominal).
3. ASSEMBLY ORDER: pulleys + belt on motor/shaft FIRST, then lower the deck onto fins + standoffs and bolt down (the fin walls close the belt corridor once the deck is on).
4. The belt passes BETWEEN the two fins, above the frame and below the deck — nothing crosses it.

SECTION ON THE BELT AXIS — LEFT SIDE (right side mirrors, +20 in z)



HARDWARE (per machine)

16x	M4 x 35 SHCS + nyloc + washers	bearing cap pairs, 4 per bore stack (2 bores x 2 plates)
8x	M5 x 16 SHCS + washer	carriage lock bolts (4 per side)
12x	M5 x 20 SHCS	fin flanges -> deck underside (10-deep flange, full heat-set)
4x	M5 x 70 SHCS + nyloc	top plate -> standoff -> deck through-bolt (one per standoff)
8x	M6 x 18 SHCS	motor flange -> carriage from below (full 8 mm engagement)
2x	M6 x 40 jackscrew + jam nut	tensioner
2x	M25x1.5 shaft locknut (KM5)	bearing preload
--	M5/M6 brass heat-set inserts	all printed bosses

5-BAR BUNG-COVER ROBOT — BASE HARDWARE	
TITLE: TENSIONER ASSEMBLY + HARDWARE	DWG NO: BCR-06 REV A
MATERIAL: -	QTY: 2 (mirrored)
SCALE 1:1 section UNITS mm THIRD ANGLE	SHEET 6 OF 6 2026-07-19